SUMMARY NOTES

DATABASE

EduForge AI • Generated Teaching Material • September 15, 2026

Executive Summary

- A **database** is a permanent, organized collection of data that can be stored, searched, and updated quickly.
- The **Database Management System (DBMS)** controls how data is stored, accessed, and protected.
- **SQL (Structured Query Language)** is the standard language for interacting with relational databases.
- Understanding databases lets students see how the apps, games, and websites they use manage huge amounts of information.

Core Concepts

- **Relational Database Basics**
- **Table (Relation)** - grid of rows (records) and columns (fields).
- **Record** - one complete set of related data (a row).
- **Field** - a single attribute of a record (a column).
- **Primary Key** - unique identifier for each record.
- **Foreign Key** - field that references a primary key in another table, creating a link.

- **SQL Essentials**
- `SELECT` - retrieve data.
- `INSERT` - add new records.
- `UPDATE` - modify existing records.
- `DELETE` - remove records.
- `WHERE` - filter rows.
- `ORDER BY` - sort results.
- `CREATE` / `DROP` - add or remove tables, indexes, etc.

- **Database Design Principles**
- **Normalization** - organizing tables to eliminate redundancy (1NF, 2NF, 3NF ...).
- **Constraints** - rules such as `NOT NULL`, `UNIQUE`, `CHECK` that keep data accurate.
- **Indexes** - accelerate searches on a column (trade-off: extra storage).
- **Transactions & ACID** - group operations so they succeed or fail together (Atomicity, Consistency, Isolation, Durability).
- **Referential Integrity** - enforced by proper primary/foreign key use, preventing orphan records.

- **Types of Database Systems**
- **Relational DBMS (RDBMS)** - tables + SQL (e.g., MySQL, PostgreSQL, Microsoft Access).
- **NoSQL** - non-tabular storage (documents, key-value, graphs) for flexible or massive data sets.

- **Special-purpose examples** - SQLite (lightweight, file-based), Microsoft Access (GUI-driven for beginners).

In-Depth Breakdown

1. History & Foundations

- Early storage: flat files, punch cards - no built-in relationships or query power.
- **Edgar F. Codd** (1970s) introduced the **relational model**, the theoretical basis for modern RDBMS.

2. Table Structure & Keys

- **Fields** have a defined data type (e.g., INTEGER, VARCHAR, DATE).
- **Primary Key** must be unique and not null; often an auto-incremented ID.
- **Foreign Key** creates a *parent-child* relationship; DBMS can enforce cascade updates/deletes.

3. SQL Syntax Highlights

Command	Purpose	Example
`SELECT`	Retrieve columns/rows	`SELECT name, grade FROM Students WHERE grade > 80;`
`INSERT`	Add a new row	`INSERT INTO Students (name, grade) VALUES ('Ana', 92);`
`UPDATE`	Change existing data	`UPDATE Students SET grade = 95 WHERE name = 'Ana';`
`DELETE`	Remove rows	`DELETE FROM Students WHERE grade < 50;`
`CREATE TABLE`	Define a new table	`CREATE TABLE Students (id INT PRIMARY KEY, name VARCHAR(50), grade INT);`
`CREATE INDEX`	Speed up look-ups	`CREATE INDEX idx_grade ON Students(grade);`

4. Normalization Steps

1. **First Normal Form (1NF)** - eliminate repeating groups; each field holds atomic values.

2. **Second Normal Form (2NF)** - remove partial dependencies; every non-key attribute depends on the whole primary key.

3. **Third Normal Form (3NF)** - remove transitive dependencies; non-key attributes depend only on the primary key.

5. Constraints & Integrity

- `NOT NULL` - field must contain a value.
- `UNIQUE` - all values in the field must be different.
- `CHECK` - custom condition (e.g., `CHECK (grade BETWEEN 0 AND 100)`).
- `FOREIGN KEY ... REFERENCES ...` - enforces referential links.

6. Transactions & ACID

- **Atomicity** - all statements succeed or none do.
- **Consistency** - database moves from one valid state to another.
- **Isolation** - concurrent transactions don't interfere.
- **Durability** - once committed, changes survive system failures.

7. NoSQL Overview (for contrast)

Category	Data Model	Typical Use
Document	JSON-like objects	Content management, flexible schemas
Key-Value	Simple key -> value pairs	Caching, session storage
Column-Family	Wide tables with dynamic columns	Big data analytics
Graph	Nodes & edges	Social networks, recommendation engines

Review Questions (15 MCQs)

1. **Which of the following best describes a primary key?**

A. A field that can contain duplicate values

B. A field that uniquely identifies each record

C. A field that stores large text blobs

D. A field that references another table

Answer: B

2. **What does the `WHERE` clause do in a SELECT statement?**

A. Sorts the result set

B. Limits the number of columns returned

C. Filters rows based on a condition

D. Creates a new table

Answer: C

3. **Which normal form eliminates repeating groups?**

A. 1NF

B. 2NF

C. 3NF

D. BCNF

Answer: A

4. **In the ACID properties, "I" stands for:**

A. Integrity

B. Isolation

C. Indexing

D. Interaction

Answer: B

5. **Which SQL command adds a new column to an existing table?**

A. `INSERT`

B. `ALTER TABLE ... ADD`

C. `CREATE COLUMN`

D. `UPDATE`

Answer: B

6. **A foreign key ensures:**

A. That a column contains only numeric data

B. That every row has a unique identifier

C. That values match a primary key in another table

D. That the table is stored in alphabetical order

Answer: C

7. **Which of the following is a NoSQL database type?**

A. MySQL

B. PostgreSQL

C. MongoDB

D. Microsoft Access

Answer: C

8. **What is the purpose of an index?**

A. To enforce referential integrity

B. To speed up data retrieval on a column

C. To store backup copies of the table

D. To automatically normalize data

Answer: B

9. **Which SQL statement would you use to remove all rows where `score` is less than 40?**

A. `DELETE FROM Scores WHERE score < 40;`

B. `REMOVE FROM Scores WHERE score < 40;`

C. `DROP Scores WHERE score < 40;`

D. `SELECT FROM Scores WHERE score < 40;`

Answer: A

10. **Which constraint prevents a column from having NULL values?**

A. `UNIQUE`

B. `PRIMARY KEY`

C. `NOT NULL`

D. `CHECK`

Answer: C

11. **In a relational database, a "record" is synonymous with:**

A. A field

B. A table

C. A row

D. An index

Answer: C

12. **Which of the following best describes the purpose of normalization?**

A. To increase the size of the database file

B. To reduce data redundancy and improve integrity

C. To encrypt all stored data

D. To create visual charts from data

Answer: B

13. **Which DBMS is most commonly embedded in mobile applications?**

A. MySQL

B. SQLite

C. Oracle

D. Microsoft Access

Answer: B

14. **What does the `ORDER BY` clause do?**

A. Deletes duplicate rows

B. Groups rows by a column

C. Sorts the result set in ascending or descending order

D. Limits the number of rows returned

Answer: C

15. **A transaction that fails after some of its statements have executed should:**

A. Commit the successful statements

B. Roll back all statements to maintain atomicity

C. Ignore the failure and continue

D. Convert to a stored procedure

Answer: B